

9.27

$$X(s) = \frac{A}{(s+1)^2+1} = \frac{16}{(s+1)^2+1} = \frac{16}{2j} \left(\frac{1}{s+1-j} - \frac{1}{s+1+j} \right)$$

$$x(t) = 16 \cdot \frac{1}{2j} (e^{(-1+j)t} - e^{(-1-j)t}) u(t) = 16 e^{-t} \sin t u(t)$$

$$\text{ROC: } \operatorname{Re}\{s\} > -1$$

9.43

(a) Causality: $\frac{dh(t)}{dt} \leftrightarrow sH(s) \checkmark$ 因果

Stable: $\frac{dh(t)}{dt} \leftrightarrow sH(s) \checkmark$ 稳定

(b) Causality: $\int_{-\infty}^t h(\tau) d\tau \leftrightarrow \frac{H(s)}{s}$ 因果

Stable: $\int_{-\infty}^t h(\tau) d\tau \leftrightarrow \frac{H(s)}{s}$ 稳定

9.56

$$\left| \int_{-\infty}^{+\infty} x(t) e^{-st} dt \right| \leq \int_{-\infty}^{+\infty} |x(t)| |e^{-(\sigma_0 + j\omega_0)t}| dt = \int_{-\infty}^{+\infty} |x(t)| e^{-\sigma_0 t} dt < \infty.$$

$$\Rightarrow X(s) \text{ 存在}$$

9.61

(a) $\phi_{xx}(t) = \int_{-\infty}^{+\infty} x(\tau) x(\tau+t) d\tau = \int_{-\infty}^{+\infty} x(\tau) h(t-\tau) d\tau.$

$$\Rightarrow x(\tau+t) = h(t-\tau)$$

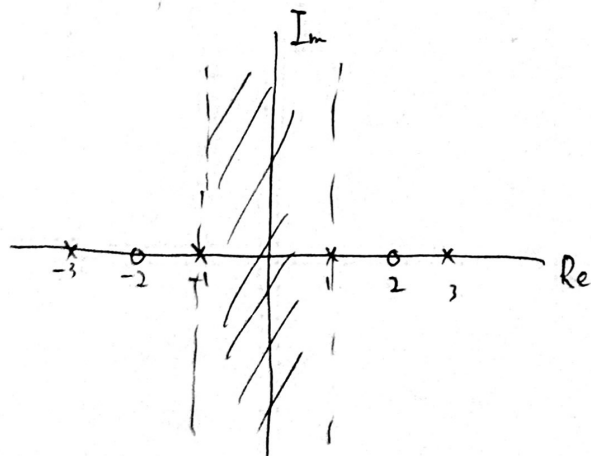
$$\Rightarrow \int_{-\infty}^{+\infty} x(\tau-t) x(\tau) d\tau \Rightarrow x(\tau-t) = h(t-\tau) \Leftrightarrow h(t) = x(-t)$$

(b) $X(s) = \int_{-\infty}^{+\infty} x(t) e^{-st} dt = \int_{-\infty}^{+\infty} x(-t) e^{st} d(-t) = \int_{-\infty}^{+\infty} x(-t) e^{st} dt$

$$\Phi_{xx}(s) = X(s) H(s) = X(s) X(-s) \quad (H(s) = \int_{-\infty}^{+\infty} x(-t) e^{-st} dt = \int_{-\infty}^{+\infty} x(t) e^{st} dt = X(-s))$$

$$\Phi_{xx}(j\omega) = X(j\omega) X(-j\omega)$$

(c)



$$ROC = |\operatorname{Re}\{s\}| < 1$$